

Microbiological analysis and whole-genome sequencing of *Neisseria gonorrhoeae* from the microbiological failures in the international, zoliflodacin, phase 3, clinical trial for treatment of uncomplicated urogenital gonorrhoea: a retrospective, genomic, observational study



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Summary

Background Zoliflodacin, a first-in-class oral bacterial, DNA gyrase (GyrB) inhibitor, showed non-inferiority to ceftriaxone combined with azithromycin in a recent large international, phase 3, randomised controlled trial for treatment of uncomplicated urogenital gonorrhoea. The aim of this study was to describe the microbiological and whole-genome sequencing (WGS) analyses of paired baseline (pre-treatment) and test-of-cure (TOC) gonococcal isolates from the zoliflodacin phase 3, randomised controlled trial to further characterise and evaluate the protocol-specified microbiological failures with zoliflodacin (n=22) or ceftriaxone and azithromycin (n=1).

Methods In this retrospective, genomic, observational study, results from antimicrobial susceptibility testing (agar dilution method) of isolates (n=960; 936 baseline isolates from 763 participants and 24 TOC isolates [23 with a paired baseline isolate in the same anatomical site] from 20 participants) collected during the zoliflodacin phase 3, randomised controlled trial done in 16 outpatient clinics in Belgium, the Netherlands, South Africa, Thailand, and the USA (Nov 6, 2019–March 16, 2023) are described. WGS analysis was performed on paired baseline and TOC isolates from participants with microbiological failures (zoliflodacin 44 isolates [19 participants]; ceftriaxone and azithromycin two isolates [one participant]), and the three baseline isolates with highest zoliflodacin minimum inhibitory concentration (MIC 0.5 mg/L).

Findings All isolates were inhibited by the same zoliflodacin concentrations (MICs ≤ 0.008 to 0.5 mg/L) as wild-type strains cultured internationally in 2013–23. In participants with a microbiological failure after zoliflodacin treatment (n=22, 19 participants), zoliflodacin MIC values for baseline and TOC isolates were similar, and resistance selection was lacking. WGS showed that five (23%) of 22 infections (95% CI 10–43 [in four participants]) of zoliflodacin microbiological failures had different strains at TOC versus baseline. In 17 zoliflodacin microbiological failures (15 participants), isolates at baseline and TOC were indistinguishable. 13 of these 17 microbiological failures, corresponding to 59% (95% CI 39–77; 13 of 22) of all zoliflodacin microbiological failures, were in urogenital or rectal sites in 11 participants and the isolates had zoliflodacin MICs less than or equal to 0.008 to 0.25 mg/L. The single microbiological failure after ceftriaxone and azithromycin treatment had different strains at TOC versus at baseline. No sequenced isolates had mutations associated with elevated zoliflodacin MICs.

Interpretation In the zoliflodacin phase 3, randomised controlled trial, 23% of the zoliflodacin microbiological failures and the single ceftriaxone and azithromycin microbiological failure had different gonococcal strains at TOC versus baseline, which suggests reinfections and not treatment failures. In addition, 59% of the zoliflodacin microbiological failures, all in anogenital sites, had no obvious microbiological explanation based on the low zoliflodacin MICs, previous pharmacodynamic studies, and no evidence of resistance selection after zoliflodacin therapy. A reinfection as the cause for these microbiological failures could not be excluded. We recommend that WGS is implemented in future randomised controlled trials for gonorrhoea treatment to further evaluate possible microbiological failures, exclude reinfections (to avoid underestimating the cure rates), and characterise antimicrobial resistance determinants.

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Introduction

Gonorrhoea is a global threat to public health, with 82 million new global cases among people aged 15–49 years estimated in 2020.¹ Effective treatment and control are seriously threatened by increasing antimicrobial resistance in *Neisseria gonorrhoeae*.^{2,3} Resistance to the last options for empirical therapy, ceftriaxone and azithromycin, has started to spread internationally.^{2–10} The international spread of ceftriaxone-resistant, mosaic *penA*-60 gonococcal strains is a major concern,^{3–10} with some of these ceftriaxone-resistant strains also expressing high-level azithromycin resistance.^{7,9,10} Novel treatment options, ideally oral ones,

are imperative, as emphasised by WHO¹¹ and other public health agencies.¹²

Zoliflodacin is an oral first-in-class spiropyrimidinetrione (bacterial topoisomerase II inhibitor) being developed for the treatment of uncomplicated gonorrhoea. Zoliflodacin has a unique mode of inhibition through the GyrB subunit of DNA gyrase, resulting in no cross-resistance with existing DNA biosynthesis inhibitors, such as fluoroquinolones.^{13,14} In an international phase 3, randomised, controlled, non-inferiority clinical trial, a single, oral 3 g dose of zoliflodacin was non-inferior to a single intramuscular dose of 500 mg ceftriaxone plus a single oral dose of

Research in context

Evidence before this study

Gonorrhoea is a global threat to public health, and effective treatment is threatened by increasing antimicrobial resistance (AMR), including resistance to the last option for empirical therapy, ceftriaxone, in *Neisseria gonorrhoeae*. Novel treatment options are imperative and two novel antimicrobials, gepotidacin and zoliflodacin, have recently finished their phase 3, randomised, controlled, non-inferiority, clinical trials for treatment of uncomplicated gonorrhoea. Zoliflodacin is an oral first-in-class spiropyrimidinetrione (bacterial topoisomerase II inhibitor). In its international phase 3 randomised controlled trial, a single, oral 3 g dose of zoliflodacin was non-inferior to ceftriaxone 500 mg plus azithromycin 1 g for the treatment of uncomplicated urogenital gonorrhoea. In the microbiologically evaluable population of the phase 3 randomised controlled trial, 22 microbiological failures with zoliflodacin across urogenital (n=15), pharyngeal (n=4), and rectal (n=3) sites in 19 participants were reported. WHO and other public health agencies recommend that “possible microbiological treatment failures” be further examined to identify indistinguishable (further support the microbiological failure) or different (suggesting reinfection) gonococcal strains using genetic typing. We searched PubMed using the terms “*Neisseria gonorrhoeae*” OR “gonorrhoea” with “whole-genome sequencing (WGS)” OR “next-generation sequencing (NGS)” AND “randomised controlled clinical (RCT)” AND “treatment” for papers that used WGS to further evaluate “possible microbiological failures” in gonorrhoea treatment randomised controlled trials, published between Jan 1, 2000, and March 1, 2025. We could not find any previous publication reporting the results of further examination of “possible microbiological failures” in any randomised controlled trial for gonorrhoea treatment that used WGS.

Added value of this study

We show that the susceptibility to zoliflodacin is high—that is, all isolates (n=960) in the phase 3 randomised controlled trial had zoliflodacin wild-type minimum inhibitory concentrations (MICs ≤0.008 to 0.5 mg/L). Furthermore, no selection of zoliflodacin

resistance or *gyrB* target mutations was identified after treatment. WGS showed that five (23%) of the 22 zoliflodacin microbiological failures (in four participants) had different gonococcal strains at test-of-cure versus baseline, suggesting reinfections and not treatment failures. In the remaining 17 zoliflodacin microbiological failures (15 participants), isolates at baseline and test-of-cure were indistinguishable. 13 of these 17 microbiological failures (13 [59%] of all 22 zoliflodacin microbiological failures) were in urogenital or rectal sites in 11 participants and the isolates had in most cases low zoliflodacin MICs. These microbiological failures had no obvious microbiological explanation based on the low zoliflodacin MICs, previous pharmacodynamic studies, and no evidence of selection of zoliflodacin resistance. Reinfection with an indistinguishable gonococcal strain that was common in the geographical setting as the cause of these microbiological failures cannot be excluded. The remaining four (18%) of the 22 (four participants) zoliflodacin microbiological failures had baseline isolates with zoliflodacin MICs in the upper end of the wild-type MIC distribution and caused infections in the pharynx. These failures might reflect that pharyngeal infections can be more difficult to cure, despite no bacterial AMR (and instead are due to pharmacokinetic factors), than urogenital and rectal infections in some patients.

Implications of all the available evidence

The in-vitro and clinical susceptibility to zoliflodacin among clinical international gonococcal isolates is high, and the zoliflodacin target (GyrB) is highly conserved. WGS is recommended to be implemented in randomised controlled trials for gonorrhoea treatment to further evaluate “possible microbiological failures”, exclude reinfections (to avoid underestimating the cure rates), and characterise AMR determinants. Continued research regarding gonococcal evolution and how antimicrobial treatment, immune response, frequency of transmission, and the microbiome among different sexual networks, gonorrhoea patients, and anatomical sites affect gonococcal evolution remains important.

1 g azithromycin (comparator) for the treatment of uncomplicated urogenital gonorrhoea (ClinicalTrials.gov NCT03959527; EudraCT 2019-000990-22).¹⁵ The zoliflodacin phase 3, randomised controlled trial was done from Nov 6, 2019, to March 16, 2023, in multiple countries, including those with a high burden of infection and gonococcal fluoroquinolone resistance, across 16 clinical sites in South Africa (n=5), Thailand (n=3), the USA (n=6), the Netherlands (n=1), and Belgium (n=1). In total, 621 participants were randomly assigned to the zoliflodacin treatment group, and 309 to the ceftriaxone and azithromycin treatment group. In the microbiologically evaluable population (including only those in the microbiological intention-to-treat populations who had a gonococcal culture result at test-of-cure [TOC]), zoliflodacin microbiological cure rates were 460 (97%) of 475, 69 (96%) of 72, and 42 (91%) of 46 for urogenital, rectal, and pharyngeal gonococcal infections, respectively. The microbiological cure rates with the comparator were 229 (100%) of 229, 31 (100%) of 31, and 22 (96%) of 23 for urogenital, rectal, and pharyngeal infections, respectively.¹⁵ Both WHO¹¹ and other international public health agencies¹² recommend that possible microbiological treatment failures are further evaluated by use of genetic typing, ie, to exclude reinfection with another gonococcal strain between treatment and TOC, and further characterise any antimicrobial-resistant gonococcal strains. For genetic typing of gonococci, including genome-based epidemiology and detection of antimicrobial resistance determinants, whole-genome sequencing (WGS) provides an ideal and accurate resolution.^{4,5,16,17}

Zoliflodacin has shown potent in-vitro activity against gonococci, including extensively drug-resistant strains. When international clinical gonococcal isolates from the past decade were examined, all isolates except one (zoliflodacin minimum inhibitory concentration [MIC] 1 mg/L)¹⁸ had MICs within the zoliflodacin wild-type MIC distribution.^{13,14,18} Isolates with elevated zoliflodacin MICs have also been selected in laboratory experiments, all with *gyrB* alleles encoding substitutions at amino acid D429 or K450 in GyrB.^{19–21} In a global WGS-based in-silico mining study, which examined 27 151 global gonococcal genomes collected from 1928 to 2021, no clinical isolates harboured GyrB K450 alterations and one (0.004%) of 27 151 harboured GyrB D429V substitution, suggesting that GyrB target-based resistance to zoliflodacin is currently extremely rare.²²

The aim of this study was to describe the microbiological analysis, with focus on susceptibility to zoliflodacin and seven additional antimicrobials for all gonococcal isolates, and WGS analysis of paired baseline (pre-treatment) and TOC (post-treatment) gonococcal isolates from the zoliflodacin international, phase 3, randomised controlled trial¹⁵ to further characterise and evaluate the protocol-specified microbiological failures (henceforth referred to as possible microbiological failures). Molecular characterisation by WGS of paired isolates obtained at baseline and

the corresponding TOC isolates from the same participants was also a pre-specified exploratory objective of the zoliflodacin phase 3 randomised controlled trial protocol.¹⁵

Methods

Study design

In this retrospective genomic observational study, we examined all baseline (day 1; appendix p 1) gonococcal isolates and TOC (day 6±2) gonococcal isolates collected throughout the zoliflodacin phase 3, randomised controlled trial.¹⁵ A flowchart of all participants recruited and randomly assigned to zoliflodacin or the comparator and details regarding study design, population, sampling, and diagnostics is provided in the main publication of the zoliflodacin phase 3, randomised controlled trial.¹⁵ Briefly, males and females aged at least 12 years with signs and symptoms consistent with urethral or endocervical gonorrhoea or a positive laboratory test confirming urogenital gonorrhoea in the preceding 14 days or a history of unprotected sexual contact in the preceding 14 days with a partner confirmed to be infected with *N gonorrhoeae*, or more than one of these, were enrolled. Clinical specimens (ie, urethral or endocervical, pharyngeal, and rectal swabs) were collected at baseline and at the TOC visit for Gram-stained microscopy (baseline specimens only), gonococcal culture and nucleic acid amplification test for *Chlamydia trachomatis* (baseline only), and *N gonorrhoeae*.²³ The protocol for the zoliflodacin phase 3, randomised controlled trial and the amendments of the protocol (ClinicalTrials NCT03959527; EudraCT 2019-000990-22) were approved by the institutional review board—ethics committee of all participating centres, and all participants provided written informed consent.¹⁵

See Online for appendix

Procedures

Antimicrobial susceptibility testing was done at JMI Laboratories, North Liberty, IA, USA. The MICs (mg/L) of zoliflodacin, ceftriaxone, cefixime, azithromycin, spectinomycin, ciprofloxacin, gentamicin, and tetracycline were established by agar dilution method as described in Clinical and Laboratory Standards Institute (CLSI) guidelines (M07-A9 and M100-S24). *N gonorrhoeae* ATCC 49226 reference strain was used for quality control. The antimicrobial MICs were interpreted by use of clinical breakpoints recommended by the EUCAST (version 15.0) and CLSI, where available. Only whole MIC doubling dilutions are reported and the ranges of zoliflodacin MICs, MIC₅₀, and MIC₉₀ for all baseline gonococcal isolates were also determined.

For all so-called possible microbiological failures in both the zoliflodacin and the ceftriaxone plus azithromycin treatment groups of the zoliflodacin phase 3, randomised controlled trial,¹⁵ paired (from the same anatomical site) baseline gonococcal isolates and TOC gonococcal isolates from the same participants (table 1) were subjected to WGS analysis. This was done to identify indistinguishable paired

	Sex assigned at birth	Race	Reported sexual orientation	Time from meal to dosing, h*	Reported unprotected sex before TOC	Site of infection	Microbiological outcome at TOC	Microbiological failure with indistinguishable isolates at TOC and baseline ^{11,12}
Participant 1†—zolidodacin								
Netherlands	Male	White	MSM	0·75	No	Pharyngeal	Failure	Yes
Netherlands	..	..	..	..	..	Urogenital	Cure	NA
Participant 2—zolidodacin								
South Africa	Male	Black or African American	HSM	0·33	No	Urogenital	Failure	Yes
Participant 3†—zolidodacin								
South Africa	Male	Black or African American	HSM	0·48	No	Urogenital	Failure	Yes
South Africa	..	..	..	..	..	Rectal	Cure	NA
Participant 4—zolidodacin								
South Africa	Male	Black or African American	HSM	0·42	No	Urogenital	Failure	Yes
Participant 5—zolidodacin								
South Africa	Male	Black or African American	HSM	1·92	No	Urogenital	Failure	Yes
Participant 6—zolidodacin								
South Africa	Male	Black or African American	HSM	0·28	No	Urogenital	Failure	Yes
South Africa	..	..	..	..	..	Rectal	Failure	Yes
Participant 7†—zolidodacin								
South Africa	Male	Black or African American	HSM	0·45	No	Urogenital	Failure	Yes
South Africa	..	..	..	..	..	Rectal	Cure	NA
Participant 8†—zolidodacin								
South Africa	Male	Black or African American	HSM	1·2	No	Urogenital	Failure	Yes
South Africa	..	..	..	..	..	Rectal	Cure	NA
Participant 9—zolidodacin								
South Africa	Male	Black or African American	HSM	0·2	No	Urogenital	Failure	Yes
Participant 10—zolidodacin								
South Africa	Male	Black or African American	HSM	1·08	No	Urogenital	Failure	Yes
Participant 11†—zolidodacin								
Thailand	Male	Asian	MSM	0·02	No	Pharyngeal	Failure	Yes
Thailand	..	..	..	..	..	Urogenital	Cure	NA
Thailand	..	..	..	..	..	Rectal	Cure	NA
Participant 12—zolidodacin								
Thailand	Male	Asian	HSM	0·02	No	Urogenital	Failure	Yes
Participant 13†—zolidodacin								
USA	Male	Black or African American	HSM	2·62	No	Pharyngeal	Failure	Yes
USA	..	..	..	..	..	Urogenital	Cure	NA
Participant 14—zolidodacin								
USA	Male	Black or African American	HSM	16·58	No	Urogenital	Failure	Yes
Participant 15—zolidodacin								
USA	Male	Black or African American	MSM	0·25	No	Urogenital	Failure	Yes
USA	..	..	..	..	..	Pharyngeal	Failure	Yes
Participant 16—zolidodacin								
Netherlands	Male	White	MSM	2·35	No	Urogenital	Failure	No
Participant 17†—zolidodacin								
South Africa	Male	Black or African American	HSM	0·53	No	Rectal	Failure	No
South Africa	..	..	..	..	..	Urogenital	Cure	NA
Participant 18†—zolidodacin								
South Africa	Male	Black or African American	HSM	0·27	Yes	Urogenital	Failure	No
South Africa	..	..	..	..	..	Rectal	Failure	No
South Africa	..	..	..	..	..	Pharyngeal	Cure	NA
Participant 19—zolidodacin								
South Africa	Female	Black or African American	HSW	1·07	No	Urogenital	Failure	No

(Table 1 continues on next page)

(Continued from previous page)

Participant 20†—ceftriaxone and azithromycin

	Sex assigned at birth	Race	Reported sexual orientation	Time from meal to dosing, h*	Reported unprotected sex before TOC	Site of infection	Microbiological outcome at TOC	Microbiological failure with indistinguishable isolates at TOC and baseline ^{11,12}
South Africa	Male	Black or African American	HSM	1.02	No	Pharyngeal	Failure	No
South Africa	..	..	..	..	..	Urogenital	Cure	NA
South Africa	..	..	..	..	..	Rectal	Cure	NA

TOC=test-of-cure. MSM=men who have sex with men (ie, male at birth and usually engaged in sexual contact with men or transgender women). NA=not applicable. HSM=Heterosexual man. HSW=heterosexual woman. *Participants who had not consumed a substantial meal within 2 h of dosing were provided with a moderate calorie-moderate fat meal; vomiting within 30 min of dosing was not reported by any of these participants. †Nine participants also had gonococcal infections in other anatomical sites (three urogenital, three rectal, two urogenital + rectal, and one pharyngeal) that were cured with zoliflodacin 3 g.

Table 1: Main demographic and clinical characteristics of the possible microbiological failures^{11,12} in the international, phase 3, randomised, controlled, clinical trial evaluating zoliflodacin 3 g single oral dose against uncomplicated gonorrhoea¹⁵

isolates at baseline and TOC (to further support a microbiological failure), or identify different strains at TOC versus baseline (suggesting a reinfection as the cause of the possible microbiological failure).^{11,12} Baseline gonococcal isolates with a zoliflodacin MIC of 0.5 mg/L were also analysed by WGS. Before WGS, all isolates were cultured on selective Modified Thayer-Martin agar plates²³ incubated at 35°C in a humid 5% CO₂-enriched atmosphere for up to 24 h. Colonies were species confirmed with Gram-stained microscopy, rapid oxidase reaction, and matrix-assisted laser desorption ionisation-time of flight mass spectrometry (MALDI-TOF MS; Bruker Daltonics, Germany).²³

For DNA extraction, single-colony subcultures were suspended in saline solution and processed by use of the QIAasympy (QIAGEN, Hilden, Germany) with the DSP DNA Midi Kit (QIAGEN). DNA library preparation for WGS was done by use of the Illumina DNA Prep kit (Illumina, San Diego, CA, USA), and paired-end sequencing (149-bp reads with a mean genome coverage of >150×) was done on a NextSeq 550 platform (Illumina), following the manufacturer's instructions.

All reads were species confirmed by use of kraken2,²⁴ potential contamination excluded, and quality controlled and trimmed with trimmomatic (version 0.39). All quality-controlled sequence reads were mapped to the gonococcal FA1090 (Genbank: AE004969.1) genome, with low-complexity regions masked with dustmasker (version 1.0.0)²⁵ with default parameters, to obtain a multiple sequence alignment (MSA). Gubbins (version 2.4.1)²⁶ was utilised and masked 163 recombinant regions or blocks (mean 10.4), generating a final recombinant-free (filtered) phylogeny. Pairwise genomic distances between paired (from the same anatomical site) baseline and TOC isolates, and all isolates, respectively, and subsequent identification of the differences between paired baseline and TOC isolates were computed by use of a recombinant-free MSA with MEGA11 (version 11.0.13). The genomic relationship between isolates was also visualised in Microreact by use of a phylogenetic tree generated in Pathogenwatch.^{4,16} A customised CLC Genomics Workbench (version 23.0.5) workflow was used to

characterise antimicrobial resistance determinants, multi-locus sequence typing (MLST) sequence types, and *N gonorrhoeae* sequence typing for antimicrobial resistance (NG-STAR) sequence types, and to generate an assembly for downstream analysis, as described.²² Mutations in the GyrB codons D429 and K450, associated with elevated zoliflodacin MICs, were identified from the WGS sequences (detection threshold: 5% single nucleotide polymorphism [SNP] frequency), as previously described.²²

Gonococcal isolates obtained at baseline and TOC were considered indistinguishable if they were assigned identical MLST and NG-STAR sequence types, contained identical alleles of antimicrobial resistance determinants, and differed by less than or equal to 20 unfiltered SNPs or less than or equal to 20 genetic events (filtered SNPs plus recombinational blocks) in their whole-genome sequences.

Statistical analysis

Wilson's method, in Stata (version 18.0), was used to calculate 95% CIs for the percentages of different groups of the possible microbiological failures.

Role of the funding source

The funders had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Results

The results of the antimicrobial susceptibility testing of the 936 baseline gonococcal isolates¹⁵ are summarised in the appendix (p 2). These 936 baseline isolates were obtained on day 1 from 763 participants from both treatment groups (ie, 741 participants had a baseline urogenital isolate and 22 additional participants were lacking a baseline urogenital isolate but had a baseline rectal or pharyngeal isolate). The baseline isolates were cultured from the urogenital tract (n=741), rectum (n=114), and pharynx (n=81; appendix p 1).¹⁵ The range of MICs, MIC₅₀, and MIC₉₀ for zoliflodacin was from less than or equal to 0.008 mg/L to 0.5 mg/L, 0.06 mg/L, and 0.12 mg/L, respectively (appendix p 2). All zoliflodacin MIC values (figure 1) were within the previously described

For more on Pathogenwatch see <https://pathogen.watch/>

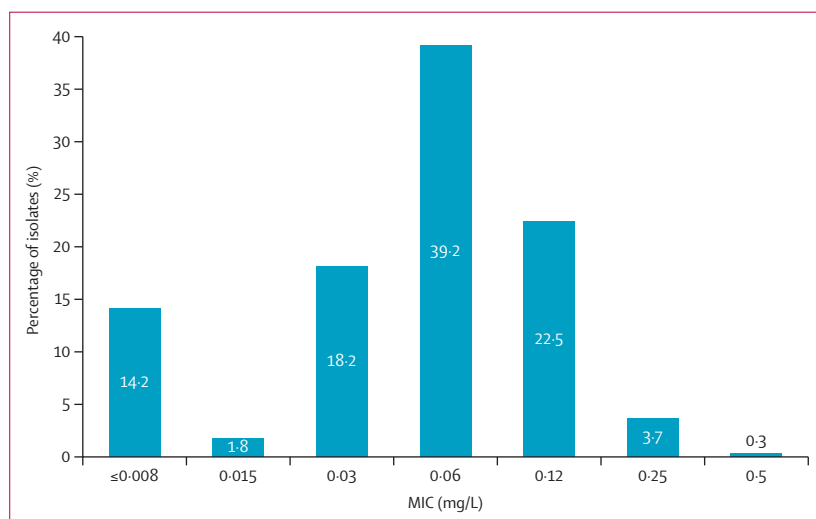


Figure 1: MIC distribution for zoliflodacin, including all the baseline (pre-treatment) *Neisseria gonorrhoeae* isolates

n=936; MIC ≤0.008 mg/L (133 of 936), MIC=0.015 mg/L (17 of 936), MIC=0.03 mg/L (170 of 936), MIC=0.06 mg/L (367 of 936), MIC=0.12 mg/L (211 of 936), MIC=0.25 mg/L (35 of 936), and MIC=0.5 mg/L (3 of 936).

MIC=minimum inhibitory concentration.

zoliflodacin MIC distribution for wild-type strains.^{13,14,18} By use of EUCAST resistance breakpoints, the proportions of isolates resistant to ceftriaxone, cefixime, azithromycin, spectinomycin, ciprofloxacin, and tetracycline were four (0.4%) of 936, 14 (2%) of 936, 56 (6%) of 936, 0 of 936, 709 (76%) of 936, and 687 (73%) of 936, respectively. By use of CLSI breakpoints, one (0.1%) of 936 and eight (0.9%) of 936 isolates were non-susceptible to ceftriaxone and cefixime, respectively, and 0 of 936, 56 (6%) of 936, 699 (75%) of 936, and 571 (61%) of 936 were resistant to spectinomycin, azithromycin, ciprofloxacin, and tetracycline, respectively (appendix p 2). The MICs of gentamicin ranged from less than or equal to 0.5 to 16 mg/L. The isolates had similar susceptibilities to zoliflodacin across anatomical sites of infection (appendix p 1). The 24 examined TOC isolates were obtained on day 6±2 from 20 participants from both treatment groups.¹⁵ 23 of these TOC isolates had a paired baseline isolate in the same anatomical site. These 23 TOC isolates were cultured from the urogenital tract (n=15), pharynx (n=5), and rectum (n=3), and were included in the subsequent WGS analysis. The 23 TOC isolates showed zoliflodacin MICs, ranging from less than or equal to 0.008 to 0.25 mg/L (tables 2, 3), within the previously described zoliflodacin MIC distribution for wild-type strains.^{13,14,18}

The demographic and clinical characteristics of participants with possible microbiological failures in the zoliflodacin treatment group (n=22, 19 participants) or ceftriaxone and azithromycin treatment group (n=1) in the zoliflodacin phase 3, randomised controlled trial¹⁵ are summarised in table 1. The failures with zoliflodacin were observed in South Africa (n=15; 278 [45%] of 621 participants randomly assigned to the zoliflodacin treatment group were in South Africa), the USA (n=3; 107 [17%] of 621), Thailand

(n=2; 181 [29%] of 621), and the Netherlands (n=2; 49 [8%] of 621), and 15 were urogenital, four pharyngeal, and three rectal.¹⁵ The single failure with ceftriaxone and azithromycin was a pharyngeal infection in South Africa. Notably, all but one participant with possible microbiological failure were male, age range was 21–55 years, and weight range was 46–97 kg; all were in the fed state at the time of dosing, and none vomited within 30 min after dosing (table 1). Paired (from the same anatomical site) baseline gonococcal isolates and TOC isolates from the same participants from all these possible microbiological failures were subjected to WGS analysis. Furthermore, for these 20 participants, baseline isolates also from anatomical sites where gonococcal infections were microbiologically cured with zoliflodacin 3 g (eight participants, nine isolates) or the comparator (one participant, two isolates) were included in the WGS analysis (table 1).

In 17 (77%) of 22 (in 15 participants; 95% CI 57–90) possible microbiological failures in the zoliflodacin treatment group, the paired baseline and TOC isolates were indistinguishable when microbiological and WGS analyses were done (table 2). Briefly, the MICs of zoliflodacin were identical (n=14) or differed by one MIC doubling dilution (n=3). The total and filtered SNP differences ranged between 0–76 and 0–9, respectively. Additionally, the MLST and NG-STAR sequence types (table 2) as well as antimicrobial resistance determinants—ie, alleles of *penA*, *mtr* operon, *penB*, 23S rRNA gene, *gyrA*, *parC*, *bla*_{TEM}, *tetM*, and *rpsJ*—were identical (appendix pp 3–4). Of note, the paired isolates from two microbiological failures (participant 3 [urogenital] and participant 11 [pharynx]) showed a distance of 76 and 37 SNPs, respectively. However, after removal of recombinational blocks in the genomes (two and five recombinational blocks, respectively), a distance of nine and zero SNPs, respectively, remained. Consequently, the paired isolates in participant 3 (urogenital) and participant 11 (pharynx) differed by 11 (nine SNPs plus two recombinational blocks) and five (zero SNPs plus five recombinational blocks) genetic events, respectively, and therefore, the isolates were considered indistinguishable. The close phylogenomic relationship between the paired baseline and TOC isolates from each participant with indistinguishable isolates at a minimum of one anatomical site (participants 1–15 in tables 1 and 2) is shown in figure 2. These 17 microbiological failures with indistinguishable isolates at baseline versus TOC occurred at the urogenital (n=12), pharyngeal (n=4), and rectal (n=1) sites. No TOC isolates had substantially elevated zoliflodacin MICs relative to the corresponding baseline isolate. However, the baseline isolates in four (24%) of 17 (95% CI 10–47) possible microbiological failures with indistinguishable baseline and TOC isolates had zoliflodacin MICs in the upper end of the zoliflodacin wild-type MIC distribution (0.12–0.5 mg/L) and caused pharyngeal infections in four participants (table 2). The baseline isolates in the remaining 13 (76%) of 17 (95% CI 53–90) possible microbiological failures with indistinguishable baseline and TOC isolates

	Country (site number in country)	Sex assigned at birth	Anatomical site	Microbiological outcome at TOC	Zoliflodacin minimum inhibitory concentration, mg/L	Total single nucleotide polymorphisms (filtered)*	MLST sequence type	NG-STAR sequence type
Participant 1†								
Baseline	Netherlands (001)	Male	Pharyngeal	Failure	0.5	1 (0)	11 422	193
TOC	..	..	..	..	0.25	..	11 422	193
Baseline	..	..	Urogenital	Cure	0.25	NA	11 422	193
Participant 2								
Baseline	South Africa (001)	Male	Urogenital	Failure	≤0.008	7 (0)	1587	427
TOC	..	..	..	..	≤0.008	..	1587	427
Participant 3†								
Baseline	South Africa (001)	Male	Urogenital	Failure	≤0.008	76 (9‡)	1587	427
TOC	..	..	..	..	≤0.008	..	1587	427
Baseline	..	..	Rectal	Cure	0.03	NA	1587	5642
Participant 4								
Baseline	South Africa (002)	Male	Urogenital	Failure	0.06	10 (0)	14 794	5643
TOC	..	..	..	..	0.06	..	14 794	5643
Participant 5								
Baseline	South Africa (002)	Male	Urogenital	Failure	0.06	0 (0)	14 794	5643
TOC	..	..	..	..	0.06	..	14 794	5643
Participant 6								
Baseline	South Africa (002)	Male	Urogenital	Failure	0.12	16 (0)	11 368	5664
TOC	..	..	..	..	0.12	..	11 368	5664
Baseline	..	..	Rectal	Failure	≤0.008	6 (0)	1587	427
TOC	..	..	..	..	≤0.008	..	1587	427
Participant 7†								
Baseline	South Africa (002)	Male	Urogenital	Failure	≤0.008	7 (0)	1587	427
TOC	..	..	..	..	≤0.008	..	1587	427
Baseline	..	..	Rectal	Cure	≤0.008	NA	1587	427
Participant 8†								
Baseline	South Africa (002)	Male	Urogenital	Failure	0.03	6 (0)	14 794	5643
TOC	..	..	..	..	0.06	..	14 794	5643
Baseline	..	..	Rectal	Cure	0.06	NA	1587	427
Participant 9								
Baseline	South Africa (004)	Male	Urogenital	Failure	0.015	20 (2‡)	18 033	1496
TOC	..	..	..	..	0.015	..	18 033	1496
Participant 10								
Baseline	South Africa (006)	Male	Urogenital	Failure	0.03	7 (3‡)	13 942	1623
TOC	..	..	..	..	0.03	..	13 942	1623
Participant 11†								
Baseline	Thailand (001)	Male	Pharyngeal	Failure	0.12	37 (0)	17 948	3549
TOC	..	..	..	..	0.12	..	17 948	3549
Baseline	..	..	Urogenital	Cure	0.06	NA	17 948	3549
Baseline	..	..	Rectal	Cure	0.12	NA	15 331	2482
Participant 12								
Baseline	Thailand (003)	Male	Urogenital	Failure	0.06	9 (0)	1587	719
TOC	..	..	..	..	0.06	..	1587	719
Participant 13†								
Baseline	USA (102)	Male	Pharyngeal	Failure	0.25	5 (3‡)	8154	3908
TOC	..	..	..	..	0.12	..	8154	3908
Baseline	..	..	Urogenital	Cure	0.06	NA	8154	3908
Participant 14								
Baseline	USA (103)	Male	Urogenital	Failure	0.06	5 (0)	15 228	2978
TOC	..	..	..	..	0.06	..	15 228	2978

(Table 2 continues on next page)

	Country (site number in country)	Sex assigned at birth	Anatomical site	Microbiological outcome at TOC	Zoliflodacin minimum inhibitory concentration, mg/L	Total single nucleotide polymorphisms (filtered)*	MLST sequence type	NG-STAR sequence type
(Continued from previous page)								
Participant 15								
Baseline	USA (103)	Male	Urogenital	Failure	0.25	12 (1‡)	9363	3966
TOC	..	..	..	..	0.25	..	9363	3966
Baseline	..	..	Pharyngeal	Failure	0.25	5 (0)	9363	3966
TOC	..	..	..	..	0.25	..	9363	3966
TOC=test-of-cure. MLST=multilocus sequence typing. NG-STAR= <i>Neisseria gonorrhoeae</i> sequence typing for antimicrobial resistance. NA=not applicable. *Comparison between the paired isolates (baseline [pretreatment] and TOC) from the same participant and anatomical site. †Six participants also had gonococcal infections in another anatomical site (two urogenital, three rectal, and one urogenital plus rectal) that were cured with zoliflodacin 3 g. ‡Single nucleotide polymorphism differences remaining after excluding recombinational blocks.								
Table 2: Possible microbiological failures^{11,12} with zoliflodacin 3 g in the international, phase 3, randomised, controlled, clinical trial,¹⁵ which had indistinguishable <i>Neisseria gonorrhoeae</i> isolates at baseline and TOC (day 6±2)								

	Country (site number in country)	Sex assigned at birth	Anatomical site	Microbiological outcome at TOC	Zoliflodacin minimum inhibitory concentration, mg/L	Total single nucleotide polymorphisms (filtered)*	MLST sequence type	NG-STAR sequence type
Participant 16								
Baseline	Netherlands (001)	Male	Urogenital	Failure	0.25	4616 (545†)	11 422	193
TOC	..	..	..	..	0.03	..	10 314	2148
Participant 17‡								
Baseline	South Africa (002)	Male	Rectal	Failure	0.06	4625 (964†)	14 601	5204
TOC	..	..	..	..	≤0.008	..	1587	5731
Baseline	..	..	Urogenital	Cure	0.015	NA	1583	5641
Participant 18‡								
Baseline	South Africa (002)	Male	Urogenital	Failure	0.03	5001 (970†)	13 942	1623
TOC	..	..	..	..	0.12	..	11 368	5664
Baseline	..	..	Rectal	Failure	≤0.008	400 (17†)	1587	427
TOC	..	..	..	..	≤0.008	..	1587	427
Baseline	..	..	Pharyngeal	Cure	≤0.008	NA	1587	427
Participant 19								
Baseline	South Africa (004)	Female	Urogenital	Failure	≤0.008	4641 (986†)	1587	427
TOC	..	..	..	..	0.015	..	1588	1632
Participant 20‡								
Baseline	South Africa (002)	Male	Pharyngeal	Failure	0.015	4569 (962†)	1583	5641
TOC	..	..	..	..	0.06	..	1588	1865
Baseline	..	..	Urogenital	Cure	0.03	NA	1583	5641
Baseline	..	..	Rectal	Cure	0.03	NA	1583	5641
TOC=test-of-cure. MLST=multilocus sequence typing. NG-STAR= <i>Neisseria gonorrhoeae</i> sequence typing for antimicrobial resistance. NA=not applicable. *Comparison between the paired isolates (baseline [pretreatment] and TOC) from the same participant and anatomical site. †Single nucleotide polymorphism differences remaining after excluding recombinational blocks. ‡Three participants also had gonococcal infections in another anatomical sites (one urogenital, one pharyngeal, and one urogenital plus rectal) that were cured with zoliflodacin 3 g or ceftriaxone 500 mg plus azithromycin 1 g.								
Table 3: Possible microbiological failures^{11,12} with zoliflodacin 3 g (participants 16–19) or ceftriaxone 500 mg plus azithromycin 1 g (participant 20) in the international, phase 3, randomised, controlled, clinical trial,¹⁵ which had different <i>Neisseria gonorrhoeae</i> strains at baseline and TOC (day 6±2)								

had mostly low zoliflodacin MIC values (≤0.008 mg/L [n=4], 0.015 mg/L [n=1], 0.03 mg/L [n=2], 0.06 mg/L [n=4], 0.125 mg/L [n=1] and, 0.25 mg/L [n=1]), and all these failures were in urogenital (n=12) or rectal (n=1) sites in 11 participants (table 2), which were all of normal weight distribution, were fed at the time of dosing, of whom none vomited within 30 min after dosing (table 1). Of note, six of these 15 participants had gonococcal infections at another

one or two anatomical sites that were a microbiological cure (seven infections in total: three urogenital and four rectal infections; table 2).

There were five possible microbiological failures that were treated with zoliflodacin 3 g (participants 16–19) and one with the comparator (participant 20) that had different paired gonococcal strains at TOC versus baseline (table 3). Briefly, the zoliflodacin MICs differed by two to three MIC

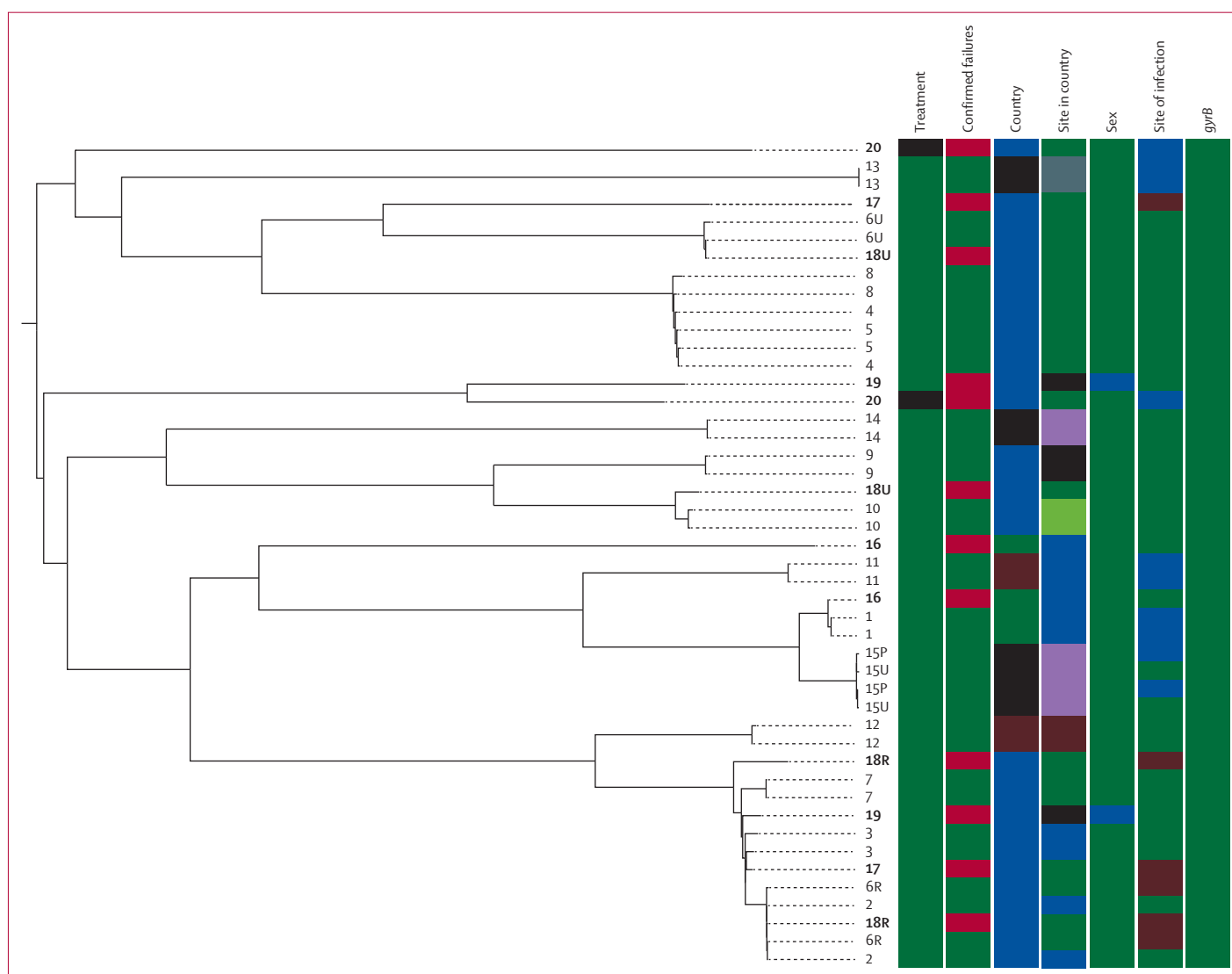


Figure 2: Phylogenomic tree representing the relationships between *Neisseria gonorrhoeae* isolates obtained at baseline (pre-treatment, n=23) and at test-of-cure (n=23) from so-called possible microbiological failures^{11,12} using zoliflodacin (n=22) or ceftriaxone and azithromycin (n=1)

Each node represents one isolate. Colour-coded columns illustrate: treatment (green: zoliflodacin, black: ceftriaxone and azithromycin); indistinguishable isolates at test-of-cure and baseline (green yes; red no); country (green the Netherlands; blue South Africa; brown Thailand; black USA); site in country (blue 001; green 002; brown 003; black 004; light green 006; grey 102; purple 103); sex (green male; blue female); site of infection (blue pharyngeal; brown rectal; green urogenital); and *gyrB* mutations (green no, blue yes). Possible microbiological failures^{11,12} that had different gonococcal strains at test-of-cure versus baseline have their isolates depicted in bold letters. The phylogenomic tree was obtained as part of the Pathogenwatch^{4,16} pipeline by use of assembled genomes.

doubling dilutions, the total and filtered SNP differences ranged between 400–5001 and 17–986, respectively, and the MLST and NG-STAR sequence types differed in five of the six microbiological failures (table 3). In addition, the alleles of *penA* and *parC* also differed in five of the six microbiological failures (appendix pp 3–5). Of note, the paired isolates from participant 18 (rectal site) showed 400 total SNPs; however, after removing 11 recombinational blocks 17 SNPs remained. Accordingly, these isolates differed in total by 28 genetic events (17 SNPs plus 11 recombinational blocks; table 3). Because it is implausible that 28 genetic events, including 11 recombinational blocks, occurred

across the genomes in 5 days (from treatment to TOC) at the rectal site, the baseline and TOC isolates were established to be different gonococcal strains (table 3). Accordingly, five (23%) of the 22 (95% CI 10–43) possible zoliflodacin microbiological failures (in four participants) had different gonococcal strains at TOC versus baseline. The phylogenomic relationship between the different strains cultured at baseline and at TOC (participants 16–20 in tables 1 and 3) is illustrated in figure 2.

The susceptibility to zoliflodacin among the whole-genome sequenced gonococcal isolates was further verified by screening for mutations associated with an elevated

zolidodacin MIC—ie, in the GyrB D429 or K450 amino acid codons.^{18–22} Despite a detection threshold capable of identifying low-frequency variants (5% SNP frequency) to rule out within-host emerging resistance, no mutations in *gyrB* resulting in substitutions in GyrB D429 or K450 were identified in any baseline or TOC isolates, including the three baseline isolates with zolidodacin MIC of 0.5 mg/L, of which two resulted in microbiological cure.

Discussion

We describe the microbiological analysis and WGS analysis of gonococcal isolates from the recent large, international, phase 3, randomised controlled trial in which zolidodacin 3 g was shown to be non-inferior to ceftriaxone 500 mg plus azithromycin 1 g dual therapy for the treatment of uncomplicated urogenital gonorrhoea.¹⁵ All the paired baseline and TOC isolates from urogenital, rectal, and pharyngeal sites had zolidodacin MICs within the wild-type distribution, and the isolates lacked mutations associated with elevated zolidodacin MICs in the target of zolidodacin (GyrB). Additionally, there was no evidence that treatment with zolidodacin selected for gonococci with a higher zolidodacin MIC or mutations in *gyrB*. We additionally phenotypically and genomically compared paired gonococcal baseline and TOC isolates from the same participants to further examine the 22 possible microbiological failures in 19 participants with zolidodacin 3 g. There was no trend for baseline isolates with a zolidodacin MIC at the upper end of the MIC distribution. Importantly, five (23%) of the 22 possible microbiological failures (in four participants) had different gonococcal strains at TOC versus baseline, which suggests reinfections and not treatment failures. Of note, the single possible microbiological failure with the comparator (pharyngeal infection) also had different gonococcal strains at TOC versus baseline, suggesting a reinfection and not a treatment failure. In contrast, in 77% (17 failures in 15 participants) of the 22 possible microbiological failures with zolidodacin, the baseline and TOC isolates were indistinguishable. The baseline isolates in four (24%) of these 17 possible microbiological failures with indistinguishable baseline and TOC isolates had zolidodacin MICs in the upper end of the zolidodacin wild-type MIC distribution and caused pharyngeal infections in four participants. Accordingly, these four microbiological failures might reflect that pharyngeal gonococcal infections can be more difficult to cure, despite no bacterial antimicrobial resistance and instead because of pharmacokinetic factors, than urogenital and rectal infections, which is the case for many antimicrobials.^{2,3,27} Nevertheless, 42 (91%) of 46 pharyngeal infections in the microbiologically evaluable populations of the zolidodacin phase 3, randomised controlled trial were cured,¹⁵ despite similar zolidodacin MICs. Notably, the baseline isolates in the remaining 13 (76%) of the 17 microbiological failures with indistinguishable baseline and TOC isolates had mostly low zolidodacin MIC values and all these microbiological failures were in urogenital (n=12) or rectal (n=1) sites in

11 participants, who were all of normal weight distribution, were fed at the time of dosing, and none of whom vomited within 30 min after dosing. Three of these 11 participants also had rectal infections that were cured. Reflecting current knowledge of efficacy and pharmacodynamics of zolidodacin 3 g single oral dose,^{21,27–29} these 13 microbiological failures in 11 participants remain unexplained. Reinfection with the original or an indistinguishable gonococcal strain—eg, from the same untreated sexual partner(s) or transmission network where few strains can cause many of the infections—remains a possibility and can not be excluded. Of note, ten of these 13 microbiological failures in urogenital and rectal sites with baseline isolates that had zolidodacin MICs of less than or equal to 0.008 to 0.12 mg/L were reported from four clinical sites in South Africa and all four microbiological failures with isolates that had zolidodacin MIC of less than or equal to 0.008 mg/L were caused by a very similar strain of MLST ST1587 and NG-STAR ST427. These MLST and NG-STAR sequence types were prevalent among gonococcal strains spreading in South Africa during the time period of the study, which has been elucidated by the WGS surveillance in the WHO Enhanced Gonococcal Antimicrobial Surveillance Programme (Golparian D, personal communication). Rapid reinfection has also been previously shown to be common in South African settings.³⁰ Enhanced understanding of the gonococcal epidemiology (eg, types of strains spreading), transmission, and risk of reinfection in these South African and other settings would be valuable. Effective partner notification and partner treatment strategies to prevent reinfection are also recommended to avoid reinfection in all settings. Of note, participants were instructed to abstain from sexual intercourse or use condoms for vaginal, anal, and oral sex, to avoid reinfection for the duration of the study. Compliance with this was also queried at TOC. Nevertheless, reliance on self-reporting of abstained or only protected sexual intercourse can be problematic. Notably, the zolidodacin phase 3, randomised controlled trial was a large and geographically and racially diverse study, in which a large proportion of the study population was from South Africa (ie, 278 [45%] of 621 the participants randomly assigned to the zolidodacin treatment group),¹⁵ and the effect of epidemiological, cultural, and socioeconomic differences on outcomes in gonorrhoea treatment randomised controlled trials would be valuable to further explore in future studies. Of note, there was only one reinfection between treatment and TOC in the comparator (ceftriaxone and azithromycin) treatment group; however, the long half-life of azithromycin (approximately 68 h) resulting in a substantially prolonged pharmacodynamic effect might have acted as prophylaxis for reinfection in this treatment group.

The limitations of this study included that reinfection with the original or an indistinguishable gonococcal strain—eg, from the same untreated sexual partner(s) or transmission network in which a few strains can cause many of the infections—can not be excluded, and

self-reporting of abstained or only protected sexual intercourse includes inherent limitations. A detailed understanding of the gonorrhoea management (including partner notification and partner treatment strategies), epidemiology, transmission, and risk of reinfection in most of the study settings was also lacking. Furthermore, when examining possible treatment failures^{11,12} by use of culture, a mixed gonococcal infection at baseline in which the one gonococcal colony selected for subculture at baseline differs from the colony selected at TOC cannot be totally excluded; however, the likelihood of this scenario is considered low. Finally, because no evidence-based genomic thresholds for identification of indistinguishable or different gonococcal isolate strains at TOC versus baseline exist, relevant thresholds were arbitrarily chosen for this study and material.

In conclusion, the susceptibility to zoliflodacin among clinical gonococcal isolates is high, and the zoliflodacin target (GyrB) is highly conserved.^{13–15,18,22} We identified five (23%, in four participants) of the 22 zoliflodacin possible microbiological failures^{11,12} in the international zoliflodacin phase 3, randomised controlled trial¹⁵ as having different gonococcal strains at baseline versus TOC, which suggests reinfections and not treatment failures. In the remaining 17 zoliflodacin possible microbiological failures (15 participants), the gonococcal isolates were indistinguishable. However, in 13 (76%) of these 17 possible microbiological failures, all in urogenital or rectal sites (11 participants), there was no obvious microbiological explanation based on the low zoliflodacin MICs and pharmacodynamic studies,^{21,27–29} and no evidence of resistance selection after zoliflodacin therapy. A reinfection as the cause for these microbiological failures could not be excluded. We recommend that WGS is implemented in future randomised controlled trials for gonorrhoea treatment to further characterise and evaluate possible microbiological failures,^{11,12} exclude reinfections with different gonococcal strains (to avoid underestimating the cure rates), and characterise antimicrobial resistance determinants. Additional understanding of how gonococcal evolution varies in different gonococcal lineages (eg, timeframes for emergence of SNPs and recombinations in different strains) and how antimicrobial treatment (and other interventions), immune response, frequency of transmission, and the microbiome among different sexual networks, gonorrhoea patients, and anatomical sites affect gonococcal genomic changes and evolution would be valuable.

Contributors

MU and AL designed, initiated, and coordinated the WGS study. DG did the DNA extractions, WGS, and bioinformatic analyses. DG and MU analysed and interpreted all the data. MU wrote the first draft of the paper. MU, DG, VE, EB, LJVP, SS, PAB, SHM, SMM, JPM, EWH, and AL read, commented on, and approved the final manuscript. All authors had full access to all the data in the study and accept responsibility for the decision to submit for publication. The first author (MU), second author (DG), and last author (AL) verified all the underlying data of the study and had final responsibility for submitting for publication.

Declaration of interests

VE, EB, LJVP, SS, and AL are current or former employees at Global Antibiotic Research and Development Partnership (GARDP), which

funded the zoliflodacin phase 3, randomised controlled trial. PAB is an employee at Antimicrobial Development Specialists, LLC. SHM, SMM, and JPM are current or former employees at Entasis Therapeutics, an affiliate of Innoviva Specialty Therapeutics and own stock. EWH has received consulting fees from GARDP and Innoviva Specialty Therapeutics. MU and DG declare no competing interests.

Data sharing

The data underlying the results of this study and related study documents (eg, study protocol, statistical analysis plan, and informed consent form) can be made available on request. Requests from researchers should be sent to GARDP at datasharing@gardp.org for consideration. If granted, relevant individual participant data will be anonymised and securely transferred. Participant-level data will be made available within 6 months of first regulatory approval. A signed data-sharing agreement might be required. The raw sequence reads will be publicly available with publication at the European Nucleotide Archive (Bioproject: PRJEB71988).

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